

SOLIDWORKS API (.NET VBA C#) — Reference (200/337): IModelDocExtension (part 6 of 8)

SOLIDWORKS API Documentation (converted from CHM)

IModelDocExtension (part 6 of 8) (Type)

SOLIDWORKS API Help

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InsertChainDimensions Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertChainDimensions Method (IModelDocExtension)

Chains dimensions for the specified entities in this drawing or sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertChainDimensions( _  
    ByVal Entities As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Entities As System.Object  
Dim value As System.Object  
  
value = instance.InsertChainDimensions(Entities)
```

C#

```
System.object InsertChainDimensions(  
    System.object Entities  
)
```

C++/CLI

```
System.Object^ InsertChainDimensions(  
& System.Object^ Entities  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Entities

Array of entities, e.g., edges, vertices, circles, and midpoints with which to chain dimensions (see **Remarks**)

Return Value

Array of [IDisplayDimensions](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.InsertChainDimensions](#).

Example

'VBA

' Preconditions:

- ' 1. Open *install_dir\samples\tutorial\advdrawings\foodprocessor.slddrw*.
- ' 2. Select the **Sheet2** tab at the bottom.
- ' 3. Open the Immediate window.

'

' Postconditions:

- ' 1. Observe the chained dimensions in **Drawing View3**.
- ' 2. Inspect the display dimensions in the Immediate window.

Option Explicit

```
Dim swApp As SldWorks.SldWorks  
Dim modDocExt As SldWorks.ModelDocExtension  
Dim Part As SldWorks.ModelDoc2
```

```

Dim selectMgr As SldWorks.SelectionMgr
Dim dimArray As Variant
Dim entityArray(3) As Object
Dim varArray As Variant
Dim myDisplayDim As SldWorks.DisplayDimension
Dim swDim As SldWorks.Dimension
Dim dimText As String
Dim k As Integer
Dim boolstatus As Boolean

Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc
    Set selectMgr = Part.SelectionManager
    Set modDocExt = Part.Extension

    Part.ClearSelection2 True
    boolstatus = Part.ActivateView("Drawing View3")
    boolstatus = Part.Extension.SelectByRay(0.107406727925462, 0.259964392021715, 375.00575, 0, 0, -1, 1.93314752083778E-03, 1, False,
0, 0)
    Set entityArray(0) = selectMgr.GetSelectedObject6(1, -1)

    boolstatus = Part.Extension.SelectByRay(0.135835367937783, 0.281001585630832, 375.00575, 0, 0, -1, 1.93314752083778E-03, 1, False,
0, 0)
    Set entityArray(1) = selectMgr.GetSelectedObject6(2, -1)

    boolstatus = Part.Extension.SelectByRay(0.140383950339754, 0.25598438241999, 375.00575, 0, 0, -1, 1.93314752083778E-03, 1, False, 0
0)
    Set entityArray(2) = selectMgr.GetSelectedObject6(3, -1)

    boolstatus = Part.Extension.SelectByRay(0.176772609555524, 0.221301441604959, 375.00275, 0, 0, -1, 1.93314752083778E-03, 1, False,
0, 0)
    Set entityArray(3) = selectMgr.GetSelectedObject6(4, -1)

    varArray = entityArray

    dimArray = modDocExt.InsertChainDimensions(varArray)

    If Not IsEmpty(dimArray) Then
        For k = 0 To UBound(dimArray)
            Set myDisplayDim = dimArray(k)
            Set swDim = myDisplayDim.GetDimension2(0)
            dimText = swDim.Value
            Debug.Print dimText
        Next k
    End If

End Sub

```

Example

[Insert Chain Dimensions \(VB.NET\)](#)
[Insert Chain Dimensions \(C#\)](#)

Remarks

The first element in Entities is the starting entity which is used to dimension the second element. The second element is used to dimension the third element, and so on. If Entities is Nothing or null, then pre-selected entities are used to create the chain dimensions.

The array of chain dimensions returned contains one fewer element than the array of Entities.

For more information, see the **SOLIDWORKS help > Detailing and Drawings > Drawings > Dimensions in Drawings > Chain Dimensions** topic.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : [InsertDatumTargetSymbol2 Method \(IModelDocExtension\)](#)

Obsolete. Superseded by [IModelDocExtension::InsertDatumTargetSymbol3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDatumTargetSymbol2( _  
    ByVal Datum1 As System.String, _  
    ByVal Datum2 As System.String, _  
    ByVal Datum3 As System.String, _  
    ByVal AreaStyle As System.Short, _  
    ByVal AreaOutside As System.Boolean, _  
    ByVal Value1 As System.Double, _  
    ByVal Value2 As System.Double, _  
    ByVal ValueStr1 As System.String, _  
    ByVal ValueStr2 As System.String, _  
    ByVal ArrowsSmart As System.Boolean, _  
    ByVal ArrowStyle As System.Short, _  
    ByVal LeaderLineStyle As System.Short, _  
    ByVal LeaderBent As System.Boolean, _  
    ByVal ShowArea As System.Boolean, _  
    ByVal ShowSymbol As System.Boolean _  
) As DatumTargetSym
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Datum1 As System.String  
Dim Datum2 As System.String  
Dim Datum3 As System.String  
Dim AreaStyle As System.Short  
Dim AreaOutside As System.Boolean  
Dim Value1 As System.Double  
Dim Value2 As System.Double  
Dim ValueStr1 As System.String  
Dim ValueStr2 As System.String  
Dim ArrowsSmart As System.Boolean  
Dim ArrowStyle As System.Short  
Dim LeaderLineStyle As System.Short  
Dim LeaderBent As System.Boolean  
Dim ShowArea As System.Boolean  
Dim ShowSymbol As System.Boolean  
Dim value As DatumTargetSym
```

```
value = instance.InsertDatumTargetSymbol2(Datum1, Datum2, Datum3, AreaStyle, AreaOutside, Value1,  
Value2, ValueStr1, ValueStr2, ArrowsSmart, ArrowStyle, LeaderLineStyle, LeaderBent, ShowArea,  
ShowSymbol)
```

C#

```
DatumTargetSym InsertDatumTargetSymbol2(  
    System.string Datum1,  
    System.string Datum2,  
    System.string Datum3,  
    System.short AreaStyle,  
    System.bool AreaOutside,  
    System.double Value1,  
    System.double Value2,  
    System.string ValueStr1,  
    System.string ValueStr2,  
    System.bool ArrowsSmart,  
    System.short ArrowStyle,  
    System.short LeaderLineStyle,  
    System.bool LeaderBent,  
    System.bool ShowArea,
```

```
        System.bool ShowSymbol  
    )
```

C++/CLI

```
DatumTargetSym^ InsertDatumTargetSymbol2(  
& System.String^ Datum1,  
& System.String^ Datum2,  
& System.String^ Datum3,  
& System.short AreaStyle,  
& System.bool AreaOutside,  
& System.double Value1,  
& System.double Value2,  
& System.String^ ValueStr1,  
& System.String^ ValueStr2,  
& System.bool ArrowsSmart,  
& System.short ArrowStyle,  
& System.short LeaderLineStyle,  
& System.bool LeaderBent,  
& System.bool ShowArea,  
& System.bool ShowSymbol  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Datum1

Datum reference string 1

Datum2

Datum reference string 2

Datum3

Datum reference string 3

AreaStyle

- 0 = point
- 1 = circle
- 2 = rectangle

AreaOutside

True to display the target area outside, false to not

Value1

Numeric datum target area diameter or width

Value2

Numeric datum target area height

ValueStr1

Value for datum target area diameter or width

ValueStr2

Value for datum target area height

ArrowsSmart

True to use smart arrows, false to not

ArrowStyle

Arrowhead style as defined by [swArrowStyle_e](#)

LeaderLineStyle

Leaderline style as defined by [swLeaderStyle_e](#)

LeaderBent

True to create a bent leader line, false to not

ShowArea

True to show the target area, false to not

ShowSymbol

True to display the target symbol, false to not

Return Value

[Datum target symbol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertDatumTargetSymbol2](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertDatumTargetSymbol3 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertDatumTargetSymbol3 Method (IModelDocExtension)

Creates a datum target symbol.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertDatumTargetSymbol3( _  
    ByVal Datum1 As System.String, _  
    ByVal Datum2 As System.String, _  
    ByVal Datum3 As System.String, _  
    ByVal AreaStyle As System.Integer, _  
    ByVal AreaOutside As System.Boolean, _  
    ByVal Value1 As System.Double, _  
    ByVal Value2 As System.Double, _  
    ByVal ValueStr1 As System.String, _  
    ByVal ValueStr2 As System.String, _  
    ByVal ArrowsSmart As System.Boolean, _  
    ByVal ArrowStyle As System.Integer, _  
    ByVal LeaderLineStyle As System.Integer, _  
    ByVal LeaderBent As System.Boolean, _  
    ByVal ShowArea As System.Boolean, _  
    ByVal ShowSymbol As System.Boolean, _  
    ByVal MoveableDatumStyle As System.Integer _  
) As DatumTargetSym
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Datum1 As System.String  
Dim Datum2 As System.String  
Dim Datum3 As System.String  
Dim AreaStyle As System.Integer  
Dim AreaOutside As System.Boolean  
Dim Value1 As System.Double  
Dim Value2 As System.Double  
Dim ValueStr1 As System.String  
Dim ValueStr2 As System.String  
Dim ArrowsSmart As System.Boolean  
Dim ArrowStyle As System.Integer  
Dim LeaderLineStyle As System.Integer  
Dim LeaderBent As System.Boolean  
Dim ShowArea As System.Boolean  
Dim ShowSymbol As System.Boolean  
Dim MoveableDatumStyle As System.Integer  
Dim value As DatumTargetSym
```

```
value = instance.InsertDatumTargetSymbol3(Datum1, Datum2, Datum3, AreaStyle, AreaOutside, Value1,  
Value2, ValueStr1, ValueStr2, ArrowsSmart, ArrowStyle, LeaderLineStyle, LeaderBent, ShowArea,  
ShowSymbol, MoveableDatumStyle)
```

C#

```
DatumTargetSym InsertDatumTargetSymbol3(  
    System.string Datum1,  
    System.string Datum2,  
    System.string Datum3,  
    System.int AreaStyle,  
    System.bool AreaOutside,  
    System.double Value1,
```

```

    System.double Value2,
    System.string ValueStr1,
    System.string ValueStr2,
    System.bool ArrowsSmart,
    System.int ArrowStyle,
    System.int LeaderLineStyle,
    System.bool LeaderBent,
    System.bool ShowArea,
    System.bool ShowSymbol,
    System.int MoveableDatumStyle
)

```

C++/CLI

```

DatumTargetSym^ InsertDatumTargetSymbol3(
& System.String^ Datum1,
& System.String^ Datum2,
& System.String^ Datum3,
& System.int AreaStyle,
& System.bool AreaOutside,
& System.double Value1,
& System.double Value2,
& System.String^ ValueStr1,
& System.String^ ValueStr2,
& System.bool ArrowsSmart,
& System.int ArrowStyle,
& System.int LeaderLineStyle,
& System.bool LeaderBent,
& System.bool ShowArea,
& System.bool ShowSymbol,
& System.int MoveableDatumStyle
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Datum1

Datum reference string 1

Datum2

Datum reference string 2

Datum3

Datum reference string 3

AreaStyle

- 0 = point
- 1 = circle
- 2 = rectangle

AreaOutside

True to display the target area outside, false to not

Value1

Numeric datum target area diameter or width

Value2

Numeric datum target area height

ValueStr1

Value for datum target area diameter or width

ValueStr2

Value for datum target area height

ArrowsSmart

True to use smart arrows, false to not

ArrowStyle

Arrowhead style as defined by [swArrowStyle_e](#)

LeaderLineStyle

Leaderline style as defined by [swLeaderStyle_e](#)

LeaderBent

True to create a bent leader line, false to not

ShowArea

True to show the target area, false to not

ShowSymbol

True to display the target symbol, false to not

MoveableDatumStyle

Moveable datum target symbol style as defined in [swMoveableDatumStyle_e](#)

Return Value

[Datum target symbol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertDatumTargetSymbol3](#).

Example

[Insert and Modify Datum Target Symbol \(VBA\)](#)
[Insert and Modify Datum Target Symbol \(VB.NET\)](#)
[Insert and Modify Datum Target Symbol \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

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InsertDeleteFace Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertDeleteFace Method (IModelDocExtension)

Inserts a DeleteFace feature.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertDeleteFace( _  
    ByVal Option As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Option As System.Integer  
Dim value As System.Boolean  
  
value = instance.InsertDeleteFace(Option)
```

C#

```
System.bool InsertDeleteFace(  
    System.int Option  
)
```

C++/CLI

```
System.bool InsertDeleteFace(  
& System.int Option  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Option

Option as defined in [swFaceDeleteOption](#) e

Return Value

True if a DeleteFace feature is inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertDeleteFace](#).

Example

[Insert and Change DeleteFace Feature \(C#\)](#)
[Insert and Change DeleteFace Feature \(VB.NET\)](#)
[Insert and Change DeleteFace Feature \(VBA\)](#)

Remarks

This is a part-level operation and only works when the model is a [part document](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDeleteFaceFeatureData Interface](#)
[IFeatureManager.EditDeleteFace Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertGeneralTableAnnotation Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertGeneralTableAnnotation Method (IModelDocExtension)

Inserts the a general table annotation in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGeneralTableAnnotation( _  
    ByVal UseAnchorPoint As System.Boolean, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal AnchorType As System.Integer, _  
    ByVal TableTemplate As System.String, _  
    ByVal Rows As System.Integer, _  
    ByVal Columns As System.Integer _  
    ) As TableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim UseAnchorPoint As System.Boolean  
Dim X As System.Double  
Dim Y As System.Double  
Dim AnchorType As System.Integer  
Dim TableTemplate As System.String  
Dim Rows As System.Integer  
Dim Columns As System.Integer  
Dim value As TableAnnotation
```

```
value = instance.InsertGeneralTableAnnotation(UseAnchorPoint, X, Y, AnchorType, TableTemplate,  
Rows, Columns)
```

C#

```
TableAnnotation InsertGeneralTableAnnotation(  
    System.bool UseAnchorPoint,  
    System.double X,  
    System.double Y,  
    System.int AnchorType,  
    System.string TableTemplate,  
    System.int Rows,
```

```
System.int Columns  
)
```

C++/CLI

```
TableAnnotation^ InsertGeneralTableAnnotation(  
& System.bool UseAnchorPoint,  
& System.double X,  
& System.double Y,  
& System.int AnchorType,  
& System.String^ TableTemplate,  
& System.int Rows,  
& System.int Columns  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseAnchorPoint

True to anchor the table by AnchorType and ignore any coordinates specified by X and Y, false to use the coordinates specified by X and Y

X

X coordinate of this table annotation

Y

Y coordinate of this table annotation

AnchorType

Type of anchor as defined in [swBOMConfigurationAnchorType_e](#); valid only if UseAnchorPoint is true and TableTemplate is empty (see **Remarks**)

TableTemplate

Path and file name of the general table template to use (see **Remarks**)

Rows

Number of rows in the table annotation; valid only if TableTemplate is empty (see **Remarks**)

Columns

Number of columns in the table annotation; valid only if TableTemplate is empty (see **Remarks**)

Return Value

[ITableAnnotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertGeneralTableAnnotation](#).

Example

[Insert General Table in Part \(VBA\)](#)

[Insert General Table in Part \(VB.NET\)](#)

[Insert General Table in Part \(C#\)](#)

Remarks

If TableTemplate is...

Then..

A valid path and file name

AnchorType, Rows, and Columns are ignored, and a general table based on TableTemplate is inserted

Empty

General table based only on all of the specified parameters, except TableTemplate, is inserted

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IDrawingDoc::InsertTableAnnotation2 Method \(\)](#)

[IModelDocExtension::InsertBomTable3 Method \(\)](#)

[IModelDocExtension::InsertTitleBlockTable Method \(\)](#)

[IModelDocExtension::GetGeneralTableAnnotation Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

InsertGeneralToleranceTableAnnotation Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertGeneralToleranceTableAnnotation Method (IModelDocExtension)

Inserts a general tolerance table annotation in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertGeneralToleranceTableAnnotation( _  
    ByVal TemplateName As System.String, _  
    ByVal X As System.Integer, _  
    ByVal Y As System.Integer _  
) As GeneralToleranceTableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim TemplateName As System.String  
Dim X As System.Integer  
Dim Y As System.Integer  
Dim value As GeneralToleranceTableAnnotation  
  
value = instance.InsertGeneralToleranceTableAnnotation(TemplateName, X, Y)
```

C#

```
GeneralToleranceTableAnnotation InsertGeneralToleranceTableAnnotation(  
    System.string TemplateName,  
    System.int X,  
    System.int Y  
)
```

C++/CLI

```
GeneralToleranceTableAnnotation^ InsertGeneralToleranceTableAnnotation(  
& System.String^ TemplateName,  
& System.int X,  
& System.int Y  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

Path and file name of the table template to use (see **Remarks**)

X

X coordinate of this table annotation

Y

Y coordinate of this table annotation

Return Value

[IGeneralToleranceTableAnnotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertGeneralToleranceTableAnnotation](#).

Example

[Insert General Tolerance Table \(VBA\)](#)

[Insert General Tolerance Table \(VB.NET\)](#)

[Insert General Tolerance Table \(C#\)](#)

Remarks

Specify TemplateName with `install_dir\lang\langlbom-standard.sldbomtbt`.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IGeneralToleranceTableFeature Interface](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertObjectFromFile Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertObjectFromFile Method (IModelDocExtension)

Inserts an OLE object from a file.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertObjectFromFile( _  
    ByVal FilePath As System.String, _  
    ByVal CreateLink As System.Boolean, _  
    ByVal Aspect As System.Integer, _  
    ByVal XPos As System.Double, _  
    ByVal YPos As System.Double, _  
    ByVal ZPos As System.Double _  
    ) As SwOLEObject
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FilePath As System.String  
Dim CreateLink As System.Boolean  
Dim Aspect As System.Integer  
Dim XPos As System.Double  
Dim YPos As System.Double  
Dim ZPos As System.Double  
Dim value As SwOLEObject
```

```
value = instance.InsertObjectFromFile(FilePath, CreateLink, Aspect, XPos, YPos, ZPos)
```

C#

```
SwOLEObject InsertObjectFromFile(  
    System.string FilePath,  
    System.bool CreateLink,  
    System.int Aspect,  
    System.double XPos,  
    System.double YPos,  
    System.double ZPos  
)
```

C++/CLI

```
SwOLEObject^ InsertObjectFromFile(  
& System.String^ FilePath,  
& System.bool CreateLink,  
& System.int Aspect,  
& System.double XPos,  
& System.double YPos,  
& System.double ZPos  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FilePath

Path and filename for the OLE object file

CreateLink

True to create an external link to the OLE object file, false to embed the OLE object on the document

Aspect

Viewing aspect of the object as defined in the DVASPECT enumeration (see **Remarks**)

XPos

x coordinate for the OLE object

YPos

y coordinate for the OLE object

ZPos

z coordinate for the OLE object

Return Value

Newly inserted [object](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertObjectFromFile](#).

Remarks

Currently, only the drawing documents use the x,y,z coordinate position. Part and assembly documents place the inserted object at the upper-right corner of the model view.

The aspect argument uses the DVASPECT enumeration, which has the following values:

- DVASPECT_CONTENT = 1
- DVASPECT_THUMBNAIL = 2
- DVASPECT_ICON = 4
- DVASPECT_DOCPRINT = 8

See the MSDN documentation for additional details about the DVASPECT enumeration.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::CreateOLEObject Method](#)

[IModelDocExtension::GetOLEObjects Method](#)

[IModelDocExtension::GetOLEObjectCount Method](#)

[IModelDocExtension::!CreateOLEObject Method](#)

[IModelDocExtension::!GetOLEObjects Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertScene Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertScene Method (IModelDocExtension)

Applies the specified scene to the model.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertScene( _  
    ByVal SceneDefinitionFile As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim SceneDefinitionFile As System.String  
Dim value As System.Boolean  
  
value = instance.InsertScene(SceneDefinitionFile)
```

C#

```
System.bool InsertScene(  
    System.string SceneDefinitionFile  
)
```

C++/CLI

```
System.bool InsertScene(  
& System.String^ SceneDefinitionFile  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SceneDefinitionFile

Fully qualified path and filename for the scene

Return Value

True if the specified scene is displayed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertScene](#).

Example

[Get Render References \(C#\)](#)
[Get Render References \(VB.NET\)](#)
[Get Render References \(VBA\)](#)

Remarks

[RealView graphics must be enabled](#) to apply and view scenes.

Call [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#) to repaint the graphics area.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::DeleteScene Method](#)
[IModelDocExtension::GetRenderCustomReferences Method](#)
[IModelDocExtension::GetRenderStockReferences Method](#)
[IConfiguration::GetScene Method \(\)](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertStackedBalloon Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertStackedBalloon Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::InsertStackedBalloon2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertStackedBalloon( _  
    ByVal Style As System.Integer, _  
    ByVal Size As System.Integer, _  
    ByVal UpperTextStyle As System.Integer, _  
    ByVal UpperText As System.String, _  
    ByVal LowerTextStyle As System.Integer, _  
    ByVal LowerText As System.String, _  
    ByVal CustomSize As System.Double, _  
    ByVal ShowQuantity As System.Boolean, _
```

```

    ByVal QuantityPlacement As System.Short, _
    ByVal QuantityDenotationText As System.String _
) As Note

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim Style As System.Integer
Dim Size As System.Integer
Dim UpperTextStyle As System.Integer
Dim UpperText As System.String
Dim LowerTextStyle As System.Integer
Dim LowerText As System.String
Dim CustomSize As System.Double
Dim ShowQuantity As System.Boolean
Dim QuantityPlacement As System.Short
Dim QuantityDenotationText As System.String
Dim value As Note

```

```

value = instance.InsertStackedBalloon(Style, Size, UpperTextStyle, UpperText, LowerTextStyle,
LowerText, CustomSize, ShowQuantity, QuantityPlacement, QuantityDenotationText)

```

C#

```

Note InsertStackedBalloon(
    System.int Style,
    System.int Size,
    System.int UpperTextStyle,
    System.string UpperText,
    System.int LowerTextStyle,
    System.string LowerText,
    System.double CustomSize,
    System.bool ShowQuantity,
    System.short QuantityPlacement,
    System.string QuantityDenotationText
)

```

C++/CLI

```

Note^ InsertStackedBalloon(
& System.int Style,
& System.int Size,
& System.int UpperTextStyle,
& System.String^ UpperText,
& System.int LowerTextStyle,
& System.String^ LowerText,
& System.double CustomSize,
& System.bool ShowQuantity,
& System.short QuantityPlacement,
& System.String^ QuantityDenotationText
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Style

Balloon style as defined in [swBalloonStyle_e](#)

Size

Balloon size as defined in [swBalloonFit_e](#)

UpperTextStyle

Style for the upper text of the balloon as defined in [swBalloonTextContent_e](#)

UpperText

Upper text in the balloon

LowerTextStyle

Style for the lower text of the balloon as defined in [swBalloonTextContent_e](#); valid for balloons only and Style must be set to swBS_SplitCirc

LowerText

Lower text in the balloon; valid for balloons only and Style must be set to swBS_SplitCirc

CustomSize

User-defined size of the balloon; Size must be set to swBF_UserDef

ShowQuantity

True to show quantity, false to not

QuantityPlacement

Placement of quantity value:

- 0 = Left
- 1 = Right
- 2 = Top

• 3 = Bottom
QuantityDenotationText
Denotation text for quantity

Return Value

[Note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.InsertStackedBalloon](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[INote.IsStackedBalloon Method](#)
[IBalloonStack Interface](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertStackedBalloon2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertStackedBalloon2 Method (IModelDocExtension)

Inserts a stack of balloons for selected items.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertStackedBalloon2( _  
    ByVal BalloonOptions As StackedBalloonOptions _  
) As Note
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim BalloonOptions As StackedBalloonOptions  
Dim value As Note
```

```
value = instance.InsertStackedBalloon2(BalloonOptions)
```

C#

```
Note InsertStackedBalloon2(  
    StackedBalloonOptions BalloonOptions  
)
```

C++/CLI

```
Note^ InsertStackedBalloon2(  
& StackedBalloonOptions^ BalloonOptions  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BalloonOptions
[IStackedBalloonOptions](#)

Return Value

[Note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.InsertStackedBalloon2](#).

Example

[Insert BOM Table and Stacked Balloon \(VBA\)](#)
[Insert BOM Table and Stacked Balloon \(VB.NET\)](#)
[Insert BOM Table and Stacked Balloon \(C#\)](#)

Remarks

To create a stack of balloons:

1. Select an item to insert the first balloon on the stack.
2. Call [IModelDocExtension::CreateStackedBalloonOptions](#) to create an [IStackedBalloonOptions](#) object.
3. Set the properties on the IStackedBalloonOptions object.
4. Call this method using the IStackedBalloonOptions object in the BalloonOptions argument.
5. Select one or more items to add one or more balloons to the stack.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[INote::IsStackedBalloon Method](#)
[IBalloonStack Interface](#)
[IModelDocExtension::InsertBOMBalloon2 Method](#)
[IDrawingDoc::AutoBalloon5 Method](#)
[INote::MakeStackedBalloon Method \(\)](#)
[INote::GetBalloonOptions Method \(\)](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertSurfaceFinishSymbol3 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertSurfaceFinishSymbol3 Method (IModelDocExtension)

Creates a surface-finish symbol based on the last selection.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertSurfaceFinishSymbol3( _  
    ByVal SymType As System.Integer, _  
    ByVal LeaderType As System.Integer, _  
    ByVal LocX As System.Double, _  
    ByVal LocY As System.Double, _  
    ByVal LocZ As System.Double, _  
    ByVal LaySymbol As System.Integer, _  
    ByVal ArrowType As System.Integer, _  
    ByVal MachAllowance As System.String, _  
    ByVal OtherVals As System.String, _  
    ByVal ProdMethod As System.String, _  
    ByVal SampleLen As System.String, _  
    ByVal MaxRoughness As System.String, _  
    ByVal MinRoughness As System.String, _  
    ByVal RoughnessSpacing As System.String _  
) As SFSymbol
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim SymType As System.Integer  
Dim LeaderType As System.Integer  
Dim LocX As System.Double  
Dim LocY As System.Double  
Dim LocZ As System.Double  
Dim LaySymbol As System.Integer  
Dim ArrowType As System.Integer  
Dim MachAllowance As System.String  
Dim OtherVals As System.String
```



```

Dim ProdMethod As System.String
Dim SampleLen As System.String
Dim MaxRoughness As System.String
Dim MinRoughness As System.String
Dim RoughnessSpacing As System.String
Dim value As SFSymbol

```

```

value = instance.InsertSurfaceFinishSymbol3(SymType, LeaderType, LocX, LocY, LocZ, LaySymbol,
ArrowType, MachAllowance, OtherVals, ProdMethod, SampleLen, MaxRoughness, MinRoughness,
RoughnessSpacing)

```

C#

```

SFSymbol InsertSurfaceFinishSymbol3(
    System.int SymType,
    System.int LeaderType,
    System.double LocX,
    System.double LocY,
    System.double LocZ,
    System.int LaySymbol,
    System.int ArrowType,
    System.string MachAllowance,
    System.string OtherVals,
    System.string ProdMethod,
    System.string SampleLen,
    System.string MaxRoughness,
    System.string MinRoughness,
    System.string RoughnessSpacing
)

```

C++/CLI

```

SFSymbol^ InsertSurfaceFinishSymbol3(
& System.int SymType,
& System.int LeaderType,
& System.double LocX,
& System.double LocY,
& System.double LocZ,
& System.int LaySymbol,
& System.int ArrowType,
& System.String^ MachAllowance,
& System.String^ OtherVals,
& System.String^ ProdMethod,
& System.String^ SampleLen,
& System.String^ MaxRoughness,
& System.String^ MinRoughness,
& System.String^ RoughnessSpacing
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SymType

Type of symbol as defined in [swSFSymType_e](#)

LeaderType

Type of leader as defined in [swLeaderStyle_e](#)

LocX

x location for symbol

LocY

y location for symbol

LocZ

z location for symbol

LaySymbol

Type of lay direction as defined in [swSFLaySym_e](#)

ArrowType

Type of arrow head as defined in [swArrowStyle_e](#)

MachAllowance

Material removal allowance

OtherVals

Other roughness values

ProdMethod

Production method and treatment

SampleLen

Sampling length

MaxRoughness

Maximum roughness

MinRoughness

Minimum roughness

RoughnessSpacing

Roughness spacing

Return Value

Newly inserted [surface finish symbol](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.InsertSurfaceFinishSymbol3](#).

Example

[Get Annotations Arrays \(VBA\)](#)
[Get Annotations Arrays \(VB.NET\)](#)
[Get Annotations Arrays \(C#\)](#)

Remarks

The SOLIDWORKS Design software uses the location parameters for this method only if the surface finish symbol has a leader leaderType != swNO_LEADER.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertTitleBlockTable Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertTitleBlockTable Method (IModelDocExtension)

Inserts a title block table in a part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertTitleBlockTable( _  
    ByVal TemplateName As System.String, _  
    ByVal X As System.Integer, _  
    ByVal Y As System.Integer _  
    ) As TitleBlockTableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim TemplateName As System.String  
Dim X As System.Integer  
Dim Y As System.Integer  
Dim value As TitleBlockTableAnnotation  
  
value = instance.InsertTitleBlockTable(TemplateName, X, Y)
```

C#

```

TitleBlockTableAnnotation InsertTitleBlockTable(
    System.string TemplateName,
    System.int X,
    System.int Y
)

```

C++/CLI

```

TitleBlockTableAnnotation^ InsertTitleBlockTable(
    & System.String^ TemplateName,
    & System.int X,
    & System.int Y
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

Fully qualified path and name of the title block table template

X

x coordinate for upper-left corner of title block table

Y

y coordinate for upper-left corner of title block table

Return Value

[Title block table annotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertTitleBlockTable](#).

Example

[Get Title Block Tables \(VBA\)](#)

[Get Title Block Tables \(VB.NET\)](#)

[Get Title Block Tables \(C#\)](#)

Remarks

Title block table templates have a filename extension of **.sldtbt**.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::InsertGeneralTableAnnotation Method\(\)](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InstallModelColorizer Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InstallModelColorizer Method (IModelDocExtension)

Installs your implemented interface of the [ISwColorContour](#) interface.

.NET Syntax

Visual Basic (Declaration)

```

Sub InstallModelColorizer( _
    ByVal PInterface As System.Object _
)

```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim PInterface As System.Object  
  
instance.InstallModelColorizer(PInterface)
```

C#

```
void InstallModelColorizer(  
    System.Object PInterface  
)
```

C++/CLI

```
void InstallModelColorizer(  
& System.Object^ PInterface  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PInterface
Pointer to your implemented interface

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InstallModelColorizer](#).

Remarks

After your implemented interface is installed, the SOLIDWORKS Design software calls [ISwColorContour::Value](#) on each graphical update. Your implemented interface must provide the corresponding color for each vertex and the value associated with each vertex. This value is displayed when the cursor is over that vertex.

Additionally, when the cursor is over the model, the SOLIDWORKS Design software calls [ISwColorContour::DisplayString](#) and passes the value to be formatted by your implemented interface, which the SOLIDWORKS Design software will then display.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::RemoveModelColorizer Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IPrintOut2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IPrintOut2 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::IPrintOut3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IPrintOut2( _  
    ByVal ArraySize As System.Integer, _  
    ByRef PageArray As System.Integer, _  
    ByVal Copies As System.Integer, _  
    ByVal Collate As System.Boolean, _  
    ByVal Printer As System.String, _  
    ByVal PrintFileName As System.String _  
)
```

Visual Basic (Usage)

 Copy Code

```

Dim instance As IModelDocExtension
Dim ArraySize As System.Integer
Dim PageArray As System.Integer
Dim Copies As System.Integer
Dim Collate As System.Boolean
Dim Printer As System.String
Dim PrintFileName As System.String

```

```
instance.IPrintOut2(ArraySize, PageArray, Copies, Collate, Printer, PrintFileName)
```

C#

```

void IPrintOut2(
    System.int ArraySize,
    ref System.int PageArray,
    System.int Copies,
    System.bool Collate,
    System.string Printer,
    System.string PrintFileName
)

```

C++/CLI

```

void IPrintOut2(
    & System.int ArraySize,
    & System.int% PageArray,
    & System.int Copies,
    & System.bool Collate,
    & System.String^ Printer,
    & System.String^ PrintFileName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize

Number of pages to print (see **Remarks**)

PageArray

- in-process, unmanaged C++: Pointer to an array of range of pages to print (see **Remarks**)

• VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Copies

Number of copies to print

Collate

True to collate copies, false to not

Printer

Name of the printer queue (see **Remarks**)

PrintFileName

Name of file to print to instead of Printer

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.IPrintOut2](#).

Remarks

This method supports printing parts, assemblies, and both single and multisheet drawings.

PageArray and **ArraySize**

For a part or assembly, the **ArraySize** and **PageArray** arguments are ignored. No dialogs or message boxes are displayed.

PageArray contains any number of pairs of values, each pair indicating the start page and end page of a range of pages to print. For example, to print sheets 1, 2, 3, 6, and 7 of a drawing, the array should contain 4 elements; 1, 3, 6, 7. This means to print pages 1-3 and 6-7. A range can be 5, 5, which means to print just page 5. If the array contains only one value, only that page is printed. If that one element is 0, then all sheets are printed.

If **ArraySize** = 0 or **PageArray** argument is NULL, the all sheets are printed.

PrintFileName

To print to a file instead to a printer, set **PrintFileName** to a non-empty name.

If the PrintFileName is empty, then the document is printed to the printer specified in Printer. If that string is empty, then the document is printed to the default printer for this document. Use [IModelDoc2::Printer](#) to get or set this value.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::PrintDirect Method](#)
[IModelDoc2::ClosePrintPreview Method](#)
[IModelDoc2::PrintPreview Method](#)
[IModelDocExtension::PrintOut2 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IPrintOut3 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IPrintOut3 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::PrintOut4](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub IPrintOut3( _  
    ByVal ArraySize As System.Integer, _  
    ByRef PageArray As System.Integer, _  
    ByVal Copies As System.Integer, _  
    ByVal Collate As System.Boolean, _  
    ByVal Printer As System.String, _  
    ByVal PrintFileName As System.String, _  
    ByVal ConvertToHighQuality As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ArraySize As System.Integer  
Dim PageArray As System.Integer  
Dim Copies As System.Integer  
Dim Collate As System.Boolean  
Dim Printer As System.String  
Dim PrintFileName As System.String  
Dim ConvertToHighQuality As System.Boolean  
  
instance.IPrintOut3(ArraySize, PageArray, Copies, Collate, Printer, PrintFileName,  
ConvertToHighQuality)
```

C#

```
void IPrintOut3(  
    System.int ArraySize,  
    ref System.int PageArray,  
    System.int Copies,  
    System.bool Collate,  
    System.string Printer,  
    System.string PrintFileName,  
    System.bool ConvertToHighQuality  
)
```

C++/CLI

```
void IPrintOut3(  
& System.int ArraySize,  
& System.int% PageArray,  
& System.int Copies,
```

```
& System.bool Collate,  
& System.String^ Printer,  
& System.String^ PrintFileName,  
& System.bool ConvertToHighQuality  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArraySize

Number of pages to print (see **Remarks**)

PageArray

- in-process, unmanaged C++: Pointer to an array of range of pages to print (see **Remarks**)

VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Copies

Number of copies to print

Collate

True to collate copies, false to not

Printer

Name of the printer queue (see **Remarks**)

PrintFileName

Name of file to print to instead of Printer

ConvertToHighQuality

True to convert to high quality, false to leave as draft quality

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IPrintOut3](#).

Remarks

This method supports printing parts, assemblies, and both single and multisheet drawings.

PageArray and **ArraySize**

For a part or assembly, the **ArraySize** and **PageArray** arguments are ignored. No dialogs or message boxes are displayed.

PageArray contains any number of pairs of values, each pair indicating the start page and end page of a range of pages to print. For example, to print sheets 1, 2, 3, 6, and 7 of a drawing, the array should contain 4 elements; 1, 3, 6, 7. This means to print pages 1-3 and 6-7. A range can be 5, 5, which means to print just page 5. If the array contains only one value, only that page is printed. If that one element is 0, then all sheets are printed.

If **ArraySize** = 0 or **PageArray** argument is NULL, the all sheets are printed.

PrintFileName

To print to a file instead to a printer, set **PrintFileName** to a non-empty name.

If the **PrintFileName** is empty, then the document is printed to the printer specified in **Printer**. If that string is empty, then the document is printed to the default printer for this document. Use [IModelDoc2::Printer](#) to get or set this value.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::PrintOut3 Method](#)

[IModelDocExtension::SaveAs Method](#)

[IModelDoc2::PrintDirect Method](#)

[IModelDoc2::PrintPreview Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRelease3rdPartyStorageStore Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IRelease3rdPartyStorageStore Method (IModelDocExtension)

Releases the specified third-party storage in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IRelease3rdPartyStorageStore( _
    ByVal SubStorageName As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim SubStorageName As System.String
Dim value As System.Boolean

value = instance.IRelease3rdPartyStorageStore(SubStorageName)
```

C#

```
System.bool IRelease3rdPartyStorageStore(
    System.string SubStorageName
)
```

C++/CLI

```
System.bool IRelease3rdPartyStorageStore(
    & System.String^ SubStorageName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SubStorageName
Name of the third-party storage

Return Value

True if the data is released, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IRelease3rdPartyStorageStore](#).

Remarks

You can use this method with [IModelDocExtension::IGet3rdPartyStorageStore](#). You must use it if you are not using the LoadFromStorageStoreNotify event. You must call this method when the call to IModelDocExtension::IGet3rdPartyStorageStore returns a NULL stream.

NOTE: If you are using serialization, then be careful with the standard MFC macros; otherwise, you can get messages like "**Unexpected File Format**" after your application is unloaded. One way of using IMPLEMENT_SERIAL is:

```
IMPLEMENT_SERIAL( CCustomAttr, CObject, VERSIONABLE_SCHEMA|0 )
```

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[DAssemblyDocEvents LoadFromStorageNotify](#)
[DDrawingDocEvents LoadFromStorageNotify](#)
[DPartDocEvents LoadFromStorageNotify](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IRemoveMaterialProperty Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IRemoveMaterialProperty Method (IModelDocExtension)

Removes material property values from this model.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IRemoveMaterialProperty( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Boolean  
  
value = instance.IRemoveMaterialProperty(Config_opt, Config_count, Config_names)
```

C#

```
System.bool IRemoveMaterialProperty(  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.bool IRemoveMaterialProperty(  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration options as defined by [swInConfigurationOpts_e](#)
Config_count
Number of configurations
Config_names
Array of configurations

Return Value

True if material property values are removed, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IRemoveMaterialProperty](#).

Remarks

This method is intended to be used on features that have a changed material property value, for example, color.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::!GetMaterialPropertyValues Method](#)
[IModelDocExtension::!SetMaterialPropertyValues Method](#)
[IModelDocExtension::SetMaterialPropertyValues Method](#)
[IModelDocExtension::RemoveMaterialProperty Method](#)

Availability

IsAbbreviatedViewActive Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsAbbreviatedViewActive Method (IModelDocExtension)

Gets or sets whether the abbreviated view is active.

.NET Syntax

Visual Basic (Declaration)

```
Function IsAbbreviatedViewActive() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
value = instance.IsAbbreviatedViewActive()
```

C#

```
System.bool IsAbbreviatedViewActive()
```

C++/CLI

```
System.bool IsAbbreviatedViewActive();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if an abbreviated view is active, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsAbbreviatedViewActive](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

IsConverted Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsConverted Method (IModelDocExtension)

Gets whether the active document was converted to the current release upon opening but has not yet been saved.

.NET Syntax

Visual Basic (Declaration)

```
Function IsConverted() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
value = instance.IsConverted()
```

C#

```
System.bool IsConverted()
```

C++/CLI

```
System.bool IsConverted();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the active document was converted to the current release upon opening but not yet been saved, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsConverted](#).

Example

Visual Basic for Applications (VBA)

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim bool As Boolean
```

```
Sub Main()
```

```
Set swApp = CreateObject("Sldworks.Application")
Set swModel = swApp.ActiveDoc
Set swModelDocExt = swModel.Extension
Debug.Print "Document converted to current release but not yet saved? " _
    & swModelDocExt.IsConverted
```

```
End Sub
```

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsDisplayPaneVisible Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsDisplayPaneVisible Method (IModelDocExtension)

Gets whether the Display Pane is visible in this active document.

.NET Syntax

Visual Basic (Declaration)

```
Function IsDisplayPaneVisible() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.Boolean
```

```
value = instance.IsDisplayPaneVisible()
```

C#

```
System.bool IsDisplayPaneVisible()
```

C++/CLI

```
System.bool IsDisplayPaneVisible();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the Display Pane is visible, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsDisplayPaneVisible](#).

Example

[Create Task Pane View Add-in \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[ISldWorks::GetActiveDisplayPane Method \(\)](#)

Availability

SOLIDWORKS 2024 SP02, Revision Number 32.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetMaterialPropertyValues Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ISetMaterialPropertyValues Method (IModelDocExtension)

Sets the material property values for this model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetMaterialPropertyValues( _  
    ByRef Material\_values As System.Double, _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Material_values As System.Double  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String
```

```
instance.ISetMaterialPropertyValues(Material_values, Config_opt, Config_count, Config_names)
```

C#

```
void ISetMaterialPropertyValues(  
    ref System.double Material\_values,  
    System.int Config\_opt,
```

```

        System.int Config\_count,
        ref System.string Config\_names
    )

```

C++/CLI

```

void ISetMaterialPropertyValues(
& System.double% Material\_values,
& System.int Config\_opt,
& System.int Config\_count,
& System.String^% Config\_names
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Material_values

Array of doubles of size 9 of material property values (see **Remarks**)

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_count

Number of configuration names

Config_names

Array of configuration names of size Config_count

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ISetMaterialPropertyValues](#).

Remarks

The material property values include color (R,G,B), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

The format of Material_values is an array of doubles:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after calling this method.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::GetMaterialPropertyValues Method](#)

[IModelDocExtension::IGetMaterialPropertyValues Method](#)

[IModelDocExtension::IRemoveMaterialProperty_Method](#)

[IModelDocExtension::RemoveMaterialProperty_Method](#)

[IModelDocExtension::SetMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsExploded Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsExploded Method (IModelDocExtension)

Gets the name of the exploded view currently shown in the model.

.NET Syntax

Visual Basic (Declaration)

```

Function IsExploded( _
    ByRef ViewName As System.String _

```

) As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ViewName As System.String  
Dim value As System.Boolean  
  
value = instance.IsExploded(ViewName)
```

C#

```
System.bool IsExploded(  
    out System.string ViewName  
)
```

C++/CLI

```
System.bool IsExploded(  
& [Out] System.String^ ViewName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewName

Name of the exploded view currently shown in the model or an empty string if the model is collapsed

Return Value

True if the specified exploded view is currently shown in the model, false if the model is currently collapsed

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsExploded](#).

Example

[Get Exploded Views for Configuration \(C#\)](#)
[Get Exploded Views for Configuration \(VB.NET\)](#)
[Get Exploded Views for Configuration \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IAssemblyDoc::AutoExplode Method \(\)](#)
[IAssemblyDoc::CreateExplodedView Method \(\)](#)
[IAssemblyDoc::GetExplodedViewConfigurationName Method \(\)](#)
[IAssemblyDoc::GetExplodedViewCount2 Method \(\)](#)
[IAssemblyDoc::GetExplodedViewNames2 Method \(\)](#)
[IAssemblyDoc::ShowExploded2 Method \(\)](#)
[IAssemblyDoc::ViewCollapseAssembly Method \(\)](#)
[IAssemblyDoc::ViewExplodeAssembly Method \(\)](#)
[IView::IsExploded Method \(\)](#)
[IView::ShowExploded Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsFutureVersion Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsFutureVersion Method (IModelDocExtension)

Gets whether this document is for a future version of SOLIDWORKS.

.NET Syntax

Visual Basic (Declaration)

Function IsFutureVersion() As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean
```

```
value = instance.IsFutureVersion()
```

C#

```
System.bool IsFutureVersion()
```

C++/CLI

```
System.bool IsFutureVersion();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the document is for a future version of SOLIDWORKS, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsFutureVersion](#).

Example

[Get Version History of Future Version Document \(VBA\)](#)
[Get Version History of Future Version Document \(VB.NET\)](#)
[Get Version History of Future Version Document \(C#\)](#)

Remarks

As of 2012 SP5, loading future file versions is supported, and [ISldWorks::OpenDoc6](#) no longer throws a [swFileLoadError_e.swFutureVersion](#) error.

Use IModelDocExtension::IsFutureVersion to determine:

- how to obtain the correct version history of a document. If a future version document is loaded in an older version of SOLIDWORKS, [IModelDoc2::VersionHistory](#) and [IModelDoc2::IVersionHistory](#) return the version history of the part template, not the version history of the future version document. To get the version history of a future version document, use [ISldWorks::VersionHistory](#) or [ISldWorks::IVersionHistory](#) to get the version history from its file.
- whether a component is for a future version of SOLIDWORKS and should be hidden in the user interface of older versions. Although they can be loaded, future version components may not be actionable in older versions of SOLIDWORKS.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2012 SP5, Revision Number 20.5

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsSamePersistentID Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsSamePersistentID Method (IModelDocExtension)

Gets whether the two specified objects have the same persistent reference IDs.

.NET Syntax

Visual Basic (Declaration)

```
Function IsSamePersistentID( _  
    ByVal PersistentID1 As System.Object, _
```

```
ByVal PersistentID2 As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim PersistentID1 As System.Object  
Dim PersistentID2 As System.Object  
Dim value As System.Integer  
  
value = instance.IsSamePersistentID(PersistentID1, PersistentID2)
```

C#

```
System.int IsSamePersistentID(  
    System.object PersistentID1,  
    System.object PersistentID2  
)
```

C++/CLI

```
System.int IsSamePersistentID(  
& System.Object^ PersistentID1,  
& System.Object^ PersistentID2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PersistentID1
Object
PersistentID2
Object

Return Value

Object equality as defined by [swObjectEquality](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsSamePersistentID](#).

Example

[Compare Selected Objects and Their Persistent Reference IDs \(VB.NET\)](#)
[Compare Selected Objects and Their Persistent Reference IDs \(VBA\)](#)
[Compare Selected Objects and Their Persistent Reference IDs \(C#\)](#)

Remarks

Use this method when your application is passed two entities that your application did not select, and your application needs to know if they are the same entity.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetObjectByPersistReference3 Method](#)
[IModelDocExtension::GetPersistReference3 Method](#)
[IModelDocExtension::GetPersistReferenceCount3 Method](#)
[IModelDocExtension::!GetPersistReference3 Method](#)
[IModelDocExtension::!GetObjectByPersistReference3 Method](#)
[ISidWorks::IsSame Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0



Obsolete. Superseded by [IModelDocExtension::IsVirtualComponent2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IsVirtualComponent() As System.String
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.String

value = instance.IsVirtualComponent()
```

C#

```
System.string IsVirtualComponent()
```

C++/CLI

```
System.String^ IsVirtualComponent();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsVirtualComponent](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsVirtualComponent2 Method (IModelDocExtension)

[See Also](#)



Obsolete. Superseded by [IModelDocExtension::IsVirtualComponent3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IsVirtualComponent2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.Object

value = instance.IsVirtualComponent2()
```

C#

```
System.object IsVirtualComponent2()
```

C++/CLI

```
System.Object^ IsVirtualComponent2();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Fully qualified paths to parent assembly components, up to and including the first non-virtual parent assembly component, if the model is a virtual component

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsVirtualComponent2](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IComponent2::IsVirtual Property](#)

Availability

SOLIDWORKS 2008 SP3, Revision Number 16.3

SOLIDWORKS API Help

[Send comments](#) on this topic.

IsVirtualComponent3 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IsVirtualComponent3 Method (IModelDocExtension)

Gets the paths to parent assembly components, up to and including the first non-virtual parent, if the model is a virtual component.

.NET Syntax

Visual Basic (Declaration)

```
Function IsVirtualComponent3(  
    ByRef PathChain As System.Object, _  
    ByRef TitleChain As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim PathChain As System.Object  
Dim TitleChain As System.Object  
Dim value As System.Boolean  
  
value = instance.IsVirtualComponent3(PathChain, TitleChain)
```

C#

```
System.bool IsVirtualComponent3(  
    out System.object PathChain,  
    out System.object TitleChain  
)
```

C++/CLI

```
System.bool IsVirtualComponent3(  
& [Out] System.Object^ PathChain,  
& [Out] System.Object^ TitleChain  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PathChain

Array of fully qualified paths to parent assembly components, up to and including the first non-virtual parent assembly component, if the model is a virtual component

TitleChain

Array of document titles; each document title corresponds to a fully qualified path in PathChain

Return Value

True if the component is virtual, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IsVirtualComponent3](#).

Example

[Get Paths and Titles of Parent of Virtual Component \(C#\)](#)
[Get Paths and Titles of Parent of Virtual Component \(VB.NET\)](#)
[Get Paths and Titles of Parent of Virtual Component \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IComponent2::IsVirtual Property](#)
[IAssemblyDoc::InsertNewVirtualAssembly Method](#)
[IComponent2::MakeVirtual Method](#)

Availability

SOLIDWORKS 2008 SP4, Revision Number 16.4

SOLIDWORKS API Help

[Send comments](#) on this topic.

JogDimension Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : JogDimension Method (IModelDocExtension)

Gets or sets whether jog points are on or off on an interactively selected linear or ordinate dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function JogDimension(  
    ByVal Jog As System.Boolean, _  
    ByVal WitnessIndex As System.Short _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Jog As System.Boolean  
Dim WitnessIndex As System.Short  
Dim value As System.Boolean  
  
value = instance.JogDimension(Jog, WitnessIndex)
```

C#

```
System.bool JogDimension(  
    System.bool Jog,  
    System.short WitnessIndex  
)
```

C++/CLI

```
System.bool JogDimension(  
& System.bool Jog,  
& System.short WitnessIndex  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Jog

True if jog points are on, false if not

WitnessIndex

Index, 0 or 1, of linear dimension extension line; ignored if an ordinate dimension

Return Value

True if operation succeeds, false if it fails

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::JogDimension](#).

Example

[Add and Offset Dimension Extension Lines Jogs \(VBA\)](#)

Remarks

This method is equivalent to clicking the right-mouse button on a linear or ordinate dimension in the SOLIDWORKS Design user-interface, selecting **Display Options** on the shortcut menu, and selecting **Jog**.

Call [IModelView::GraphicsRedraw](#) after setting jog points to redraw the graphics area.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDisplayDimension::GetJogParameters Method](#)
[IDisplayDimension::SetJogParameters Method](#)
[IDisplayDimension::Jogged Property](#)
[IModelDocExtension::AddOrdinateDimension Method](#)
[IDisplayDimension::AutoJogOrdinate Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

LinkedDisplayState Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : LinkedDisplayState Property (IModelDocExtension)

Gets or sets whether a display state is linked in this part.

.NET Syntax

Visual Basic (Declaration)

```
Property LinkedDisplayState As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean
```

```
instance.LinkedDisplayState = value
```

```
value = instance.LinkedDisplayState
```

C#

```
System.bool LinkedDisplayState {get; set;}
```

C++/CLI

```
property System.bool LinkedDisplayState {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if linked, false if not linked

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::LinkedDisplayState](#).

Example

[Create, Unlink, and Purge Display States in Part \(VB.NET\)](#)
[Create, Unlink, and Purge Display States in Part \(VBA\)](#)
[Create, Unlink, and Purge Display States in Part \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IConfiguration::ApplyDisplayState Method](#)
[IConfiguration::CreateDisplayState Method](#)
[IConfiguration::DeleteDisplayState Method](#)
[IConfiguration::GetDisplayStates Method](#)
[IConfiguration::GetDisplayStatesCount Method](#)
[IConfiguration::CopyDisplayStateFromConfiguration Method](#)
[IModelDocExtension::PurgeDisplayState Method](#)
[IConfigurationManager::LinkDisplayStatesToConfigurations Property \(\)](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ListExternalFileReferences Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ListExternalFileReferences Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::ListExternalFileReferences2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ListExternalFileReferences( _  
    ByRef ModelPathName As System.Object, _  
    ByRef ComponentPathName As System.Object, _  
    ByRef Feature As System.Object, _  
    ByRef DataType As System.Object, _  
    ByRef Status As System.Object, _  
    ByRef RefEntity As System.Object, _  
    ByRef FeatCom As System.Object, _  
    ByRef ConfigOption As System.Integer, _  
    ByRef ConfigName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ModelPathName As System.Object  
Dim ComponentPathName As System.Object  
Dim Feature As System.Object  
Dim DataType As System.Object  
Dim Status As System.Object  
Dim RefEntity As System.Object  
Dim FeatCom As System.Object  
Dim ConfigOption As System.Integer  
Dim ConfigName As System.String
```

```
instance.ListExternalFileReferences(ModelPathName, ComponentPathName, Feature, DataType, Status,  
RefEntity, FeatCom, ConfigOption, ConfigName)
```

C#

```
void ListExternalFileReferences(  
    out System.object ModelPathName,  
    out System.object ComponentPathName,  
    out System.object Feature,  
    out System.object DataType,  
    out System.object Status,  
    out System.object RefEntity,
```

```

    out System.object FeatCom,
    out System.int ConfigOption,
    out System.string ConfigName
)

```

C++/CLI

```

void ListExternalFileReferences(
& [Out] System.Object^ ModelPathName,
& [Out] System.Object^ ComponentPathName,
& [Out] System.Object^ Feature,
& [Out] System.Object^ DataType,
& [Out] System.Object^ Status,
& [Out] System.Object^ RefEntity,
& [Out] System.Object^ FeatCom,
& [Out] System.int ConfigOption,
& [Out] System.String^ ConfigName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ModelPathName

Array of path names of documents

ComponentPathName

Array of path names of referenced components

Feature

Array of in-context items (sketches, features, and so on)

DataType

Array of the type of data used to create the items (converted edge or face, converted or offset sketch entity, body, and so on)

Status

Array of the statuses of the external references as defined in [swExternalReferenceStatus_e](#)

RefEntity

Array of the actual items being used and the names of the documents that contain the items

FeatCom

Array of the names of the components in which the affected features exist; this information is only displayed when one or more RefEntity is in a different component in an assembly and does not apply to derived parts

ConfigOption

Configuration options as defined by [swExternalFileReferencesConfig_e](#)

ConfigName

Name of the configuration when configOption is swExternalFileReferencesNamedConfig

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ListExternalFileReferences](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::ListExternalFileReferences Method](#)

[IModelDocExtension::ListExternalFileReferencesCount Method](#)

[IModelDocExtension::UpdateExternalFileReferences Method](#)

[IModelDoc2::BreakAllExternalReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)

[IModelDoc2::LockAllExternalReferences Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ListExternalFileReferences2 Method (IModelDocExtension)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ListExternalFileReferences2 Method (IModelDocExtension)

Gets the specified external file reference information for this part or assembly.

.NET Syntax

```

Sub ListExternalFileReferences2( _
    ByRef ModelPathName As System.Object, _
    ByRef ComponentPathName As System.Object, _
    ByRef Feature As System.Object, _
    ByRef DataType As System.Object, _
    ByRef Status As System.Object, _
    ByRef RefEntity As System.Object, _
    ByRef FeatCom As System.Object, _
    ByRef ConfigOptions As System.Object, _
    ByRef ConfigNames As System.Object _
)

```



```

Dim instance As IModelDocExtension
Dim ModelPathName As System.Object
Dim ComponentPathName As System.Object
Dim Feature As System.Object
Dim DataType As System.Object
Dim Status As System.Object
Dim RefEntity As System.Object
Dim FeatCom As System.Object
Dim ConfigOptions As System.Object
Dim ConfigNames As System.Object

```

```

instance.ListExternalFileReferences2(ModelPathName, ComponentPathName, Feature, DataType, Status,
RefEntity, FeatCom, ConfigOptions, ConfigNames)

```

```

void ListExternalFileReferences2(
    out System.object ModelPathName,
    out System.object ComponentPathName,
    out System.object Feature,
    out System.object DataType,
    out System.object Status,
    out System.object RefEntity,
    out System.object FeatCom,
    out System.object ConfigOptions,
    out System.object ConfigNames
)

```

```

void ListExternalFileReferences2(
    & [Out] System.Object^ ModelPathName,
    & [Out] System.Object^ ComponentPathName,
    & [Out] System.Object^ Feature,
    & [Out] System.Object^ DataType,
    & [Out] System.Object^ Status,
    & [Out] System.Object^ RefEntity,
    & [Out] System.Object^ FeatCom,
    & [Out] System.Object^ ConfigOptions,
    & [Out] System.Object^ ConfigNames
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ModelPathName

Array of path names of externally referenced model documents

ComponentPathName

Array of path names of externally referenced assembly components

Feature

Array of in-context items (e.g., FeatureManager design tree sketches, features, and so on) that externally reference ModelPathName and ComponentPathName elements

DataType

Array of data types used to create RefEntity (e.g., arc, line, sketch plane, converted edge or face, converted or offset sketch entity, body, and so on)

Status

Array of statuses of the external references as defined by [swExternalReferenceStatus_e](#)

RefEntity

Array of actual items being used (FeatureManager design tree entities in the Feature parameter's external references) and the names of the documents containing the items

FeatCom

Array of names of the components in which the affected Features exist; valid only if one or more RefEntity is in a different component in an assembly; not valid for derived parts

ConfigOptions

Array of configuration options as defined by [swExternalFileReferencesConfig_e](#)

ConfigNames

Array of configuration names when corresponding element of ConfigOptions array contains swExternalFileReferencesNamedConfig, empty strings otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ListExternalFileReferences2](#).

Example

[Get External References \(VBA\)](#)
[Get External References \(VB.NET\)](#)
[Get External References \(C#\)](#)

Remarks

This method conveniently collects the same information that is displayed when you right-click on a feature or component with { -> } at the end of its name in the FeatureManager design tree and select **External References...** from the context menu.

The External References dialog displays the assembly or part name of the external reference, configuration if applicable, name of the feature in this document that has an external reference, status of the external reference, the entity in the external reference to which this feature or component points, and the data type of the external reference entity.

For more information, see the **SOLIDWORKS Design user-interface help > Assemblies > Top-Down Design > External References > External References Dialog Box**.

The elements of the arrays returned by this method map one to one.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::ListExternalFileReferencesCount Method \(\)](#)
[IModelDocExtension::UpdateExternalFileReferences Method \(\)](#)
[IModelDoc2::BreakAllExternalReferences Method \(\)](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method \(\)](#)
[IModelDoc2::LockAllExternalReferences Method \(\)](#)

Availability

SOLIDWORKS 2021 SP01, Revision Number 29.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

ListExternalFileReferencesCount Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ListExternalFileReferencesCount Method (IModelDocExtension)

Gets the number of external references on this part or assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function ListExternalFileReferencesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Integer
```

```
value = instance.ListExternalFileReferencesCount()
```

C#

```
System.int ListExternalFileReferencesCount()
```

C++/CLI

```
System.int ListExternalFileReferencesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of external references

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ListExternalFileReferencesCount](#).

Example

[Get External References \(VBA\)](#)
[Get External References \(VB.NET\)](#)
[Get External References \(C#\)](#)

Remarks

Call this method before calling [IModelDocExtension::ListExternalFileReferences](#) to get the number of external references on the part.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)
[IModelDoc2::BreakAllExternalReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)
[IModelDoc2::LockAllExternalReferences Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

LoadDraftingStandard Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : LoadDraftingStandard Method (IModelDocExtension)

Loads a custom drafting standard from a file.

.NET Syntax

Visual Basic (Declaration)

```
Function LoadDraftingStandard( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.LoadDraftingStandard(FileName)
```

C#

```
System.bool LoadDraftingStandard(  
    System.string FileName  
)
```

C++/CLI

```
System.bool LoadDraftingStandard(  
& System.String^ FileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and filename of the drafting standard to load

Return Value

True if the specified drafting standard is loaded, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::LoadDraftingStandard](#).

Example

[Change Drafting Standard to Custom \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::CopyDraftingStandard Method](#)

[IModelDocExtension::DeleteDraftingStandard Method](#)

[IModelDocExtension::GetDraftingStandardNames Method](#)

[IModelDocExtension::RenameDraftingStandard Method](#)

[IModelDocExtension::SaveDraftingStandard Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Make3DExperienceCompatible Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : Make3DExperienceCompatible Method (IModelDocExtension)

Makes the current model compatible with [SOLIDWORKS Design connected to the 3DEXPERIENCE platform](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub Make3DExperienceCompatible( _  
    ByVal IncludeComponents As System.Boolean, _  
    ByRef Result As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim IncludeComponents As System.Boolean  
Dim Result As System.Boolean
```

```
instance.Make3DExperienceCompatible(IncludeComponents, Result)
```

C#

```
void Make3DExperienceCompatible(
    System.bool IncludeComponents,
    out System.bool Result
)
```

C++/CLI

```
void Make3DExperienceCompatible(
    & System.bool IncludeComponents,
    & [Out] System.bool Result
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncludeComponents

True to make components compatible with 3DExperience, false to not

Result

True if model successfully made compatible with 3DExperience, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Make3DExperienceCompatible](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2022 SP01, Revision Number 30.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

MoveDecal Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : MoveDecal Method (IModelDocExtension)

Moves the decal up or down in the list of decals applied to the model.

.NET Syntax

Visual Basic (Declaration)

```
Function MoveDecal( _
    ByVal DecalID As System.Integer, _
    ByVal MoveUp As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension
Dim DecalID As System.Integer
Dim MoveUp As System.Boolean
Dim value As System.Boolean

value = instance.MoveDecal(DecalID, MoveUp)
```

C#

```
System.bool MoveDecal(
    System.int DecalID,
    System.bool MoveUp
)
```

C++/CLI



```
System.bool MoveDecal(
& System.int DecalID,
& System.bool MoveUp
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DecalID
[Decal ID](#)
MoveUp

True to move the decal up in the list of decals, false to move the decal down in the list of decals

Return Value

True if the decal is moved up or down in the list of decals, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::MoveDecal](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDecal Interface](#)
[IModelDocExtension::AddDecal Method](#)
[IModelDocExtension::CreateDecal Method](#)
[IModelDocExtension::DeleteAllDecals Method](#)
[IModelDocExtension::DeleteDecal Method](#)
[IModelDocExtension::GetDecal Method](#)
[IModelDocExtension::GetDecals Method](#)
[IModelDocExtension::GetDecalsCount Method](#)
[IModelDocExtension::HideDecal Method](#)
[IModelDocExtension::UpdateRenderMaterialsInSceneGraph Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

MoveOrCopy Method (IModelDocExtension)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : MoveOrCopy Method (IModelDocExtension)

Moves and optionally copies the selected sketch entities or annotations.

.NET Syntax

Visual Basic (Declaration)

```
Sub MoveOrCopy( _
    ByVal Copy As System.Boolean, _
    ByVal NumCopies As System.Integer, _
    ByVal KeepRelations As System.Boolean, _
    ByVal BaseX As System.Double, _
    ByVal BaseY As System.Double, _
    ByVal BaseZ As System.Double, _
    ByVal DestX As System.Double, _
    ByVal DestY As System.Double, _
    ByVal DestZ As System.Double _
)
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension
Dim Copy As System.Boolean
Dim NumCopies As System.Integer
Dim KeepRelations As System.Boolean
Dim BaseX As System.Double
Dim BaseY As System.Double
```



```
Dim BaseZ As System.Double
Dim DestX As System.Double
Dim DestY As System.Double
Dim DestZ As System.Double
```

```
instance.MoveOrCopy(Copy, NumCopies, KeepRelations, BaseX, BaseY, BaseZ, DestX, DestY, DestZ)
```

C#

```
void MoveOrCopy(
    System.bool Copy,
    System.int NumCopies,
    System.bool KeepRelations,
    System.double BaseX,
    System.double BaseY,
    System.double BaseZ,
    System.double DestX,
    System.double DestY,
    System.double DestZ
)
```

C++/CLI

```
void MoveOrCopy(
    & System.bool Copy,
    & System.int NumCopies,
    & System.bool KeepRelations,
    & System.double BaseX,
    & System.double BaseY,
    & System.double BaseZ,
    & System.double DestX,
    & System.double DestY,
    & System.double DestZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Copy

True to copy the sketch entities or annotations, false to not

NumCopies

Number of copies

KeepRelations

True to keep sketch relations, false to not

BaseX

X coordinate of the base point from which to move the sketch entities or annotations

BaseY

Y coordinate of the base point from which to move the sketch entities or annotations

BaseZ

Z coordinate of the base point from which to move the sketch entities or annotations

DestX

X coordinate of the destination point to which to move the sketch entities or annotations

DestY

Y coordinate of the destination point to which to move the sketch entities or annotations

DestZ

Z coordinate of the destination point to which to move the sketch entities or annotations

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::MoveOrCopy](#).

Example

[Move Copy Sketch Entities \(C#\)](#)

[Move Copy Sketch Entities \(VB.NET\)](#)

[Move Copy Sketch Entities \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::RotateOrCopy Method](#)

[IModelDocExtension::ScaleOrCopy Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MultiSelect Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : MultiSelect Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::MultiSelect2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Function MultiSelect( _  
    ByVal Objects As System.Object, _  
    ByVal AppendFlag As System.Boolean, _  
    ByVal Data As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Objects As System.Object  
Dim AppendFlag As System.Boolean  
Dim Data As System.Object  
Dim value As System.Integer  
  
value = instance.MultiSelect(Objects, AppendFlag, Data)
```

C#

```
System.int MultiSelect(  
    System.object Objects,  
    System.bool AppendFlag,  
    System.object Data  
)
```

C++/CLI

```
System.int MultiSelect(  
& System.Object^ Objects,  
& System.bool AppendFlag,  
& System.Object^ Data  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Objects

Array of selectable objects:

- can be the same type of objects (e.g., an array of faces or an array of edges) or different types of objects (e.g., an array of entities of faces and edges)
- if an object is not selected, then it is ignored

AppendFlag

True to append the objects to the selection list, false to replace the current selection list with these objects

Data

[ISelectData](#) object, Nothing, or null

Return Value

Number of objects selected in the model

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::MultiSelect](#).

Remarks

If this now obsolete method is used to populate a selection box in a PropertyManager page, then [IPropertyManagerPage2Handler9::OnSubmitSelection](#) only fires when set to true and [ISelectData::Mark](#) is set to 0.

Use [IModelDocExtension::MultiSelect2](#) if you need IPropertyManagerPage2Handler9::OnSubmitSelection to fire when set to true and ISelectData::Mark is set to either a 0 or non-0 value.

The [Callout](#) property of the [ISelectData](#) object is not used by this method.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::IModelDocSelectByID2 Method](#)
[IModelDocExtension::SelectByID2 Method](#)
[ISelectionMgr Interface](#)
[IModelDocExtension::SelectAll Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MultiSelect2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : MultiSelect2 Method (IModelDocExtension)

Selects multiple objects and returns the number of objects selected in the model.

.NET Syntax

Visual Basic (Declaration)

```
Function MultiSelect2( _  
    ByVal Objects As System.Object, _  
    ByVal AppendFlag As System.Boolean, _  
    ByVal Data As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Objects As System.Object  
Dim AppendFlag As System.Boolean  
Dim Data As System.Object  
Dim value As System.Integer
```

```
value = instance.MultiSelect2(Objects, AppendFlag, Data)
```

C#

```
System.int MultiSelect2(  
    System.object Objects,  
    System.bool AppendFlag,  
    System.object Data  
)
```

C++/CLI

```
System.int MultiSelect2(  
& System.Object^ Objects,  
& System.bool AppendFlag,  
& System.Object^ Data  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Objects

Array of selectable objects:

- can be the same type of objects (e.g., an array of faces or an array of edges) or different types of objects (e.g., an array of entities of faces and edges)

- if an object is not selected, then it is ignored

AppendFlag

True to append the objects to the selection list, false to replace the current selection list with these objects

Data

[ISelectData](#) object, Nothing, or null

Return Value

Number of objects selected in the model

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::MultiSelect2](#).

Example

[Select Multiple Objects for Selection Boxes \(C#\)](#)
[Select Multiple Objects for Selection Boxes \(VB.NET\)](#)
[Select Multiple Objects for Selection Boxes \(VBA\)](#)
[Specify IGES Levels and Values, Then Import IGES File \(C#\)](#)
[Specify IGES Levels and Values, Then Import IGES File \(VB.NET\)](#)
[Specify IGES Levels and Values, Then Import IGES File \(VBA\)](#)
[Multiselect Same and Different Objects \(C#\)](#)
[Multiselect Same and Different Objects \(VB.NET\)](#)
[Multiselect Same and Different Objects \(VBA\)](#)

Remarks

If using this method to populate...	Then...
A selection box in a PropertyManager page	IPropertyManagerPage2Handler9::OnSubmitSelection fires when set to true and ISelectData::Mark is set to either a 0 or a non-0 value. NOTE: IPropertyManagerPage2Handler9::OnSubmitSelection only fires when set to true and ISelectData::Mark is set to 0 if using either of the now obsolete ModelDocExtension::MultiSelect and IModelDocExtension::IMultiSelect methods.
Multiple selection boxes with different multiple selections in a PropertyManager page	Set: • each selection box's IPropertyManagerPageSelectionbox::Mark to a different power of two; for example, 1, 2, 4, 8, etc. (Setting a selection box's mark to 0 causes all selections to appear in that selection box and the active selection box.) • ISelectData::Mark and each selection box's IPropertyManagerPageSelectionbox::Mark to the same value. See the examples for details.

NOTE: If an object is already selected with a mark of x and you attempt to select the same object again using this method with AppendFlag = true, then that object remains selected with a mark with x. Reselecting a previously selected object does not assign a new mark value.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SelectById2 Method](#)
[IModelDocExtension::SelectAll Method](#)
[ISelectionMgr Interface](#)

Availability

SolidWorks 2012 SP04, Revision Number 20.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

NeedsRebuild Property (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : NeedsRebuild Property (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::NeedsRebuild2](#).

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property NeedsRebuild As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.Boolean
```

```
value = instance.NeedsRebuild
```

C#

```
System.bool NeedsRebuild {get;}
```

C++/CLI

```
property System.bool NeedsRebuild {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if document needs to be rebuilt, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::NeedsRebuild](#).

Remarks

Because this now obsolete property does not recognize frozen features, use [IModelDocExtension::NeedsRebuild2](#), which does.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2004 SP3, Revision Number 12.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

NeedsRebuild2 Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : NeedsRebuild2 Property (IModelDocExtension)

Gets whether the model document needs to be rebuilt.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property NeedsRebuild2 As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Integer  
  
value = instance.NeedsRebuild2
```

C#

```
System.int NeedsRebuild2 {get;}
```

C++/CLI

```
property System.int NeedsRebuild2 {  
    System.int get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Rebuild status of model document as defined in [swModelRebuildStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::NeedsRebuild2](#).

Example

[Move Freeze Bar \(C#\)](#)
[Move Freeze Bar \(VB.NET\)](#)
[Move Freeze Bar \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::EditRebuild3 Method](#)
[IModelDoc2::ForceRebuild3 Method](#)
[IModelDoc2::Rebuild Method](#)
[IConfiguration::AddRebuildSaveMark](#)
[IConfiguration::NeedsRebuild](#)
[IModelDocExtension::ShowPartRebuildIndicators Property](#)
[IModelDocExtension::EditRebuildAll Method](#)
[IModelDocExtension::ForceRebuildAll Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

PrintOut Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PrintOut Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::PrintOut2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub PrintOut( _  
    ByVal FromPage As System.Integer, _  
    ByVal ToPage As System.Integer, _  
    ByVal Copies As System.Integer, _  
    ByVal Collate As System.Boolean, _  
    ByVal Printer As System.String, _  
    ByVal PrintFileName As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FromPage As System.Integer  
Dim ToPage As System.Integer  
Dim Copies As System.Integer  
Dim Collate As System.Boolean  
Dim Printer As System.String  
Dim PrintFileName As System.String  
  
instance.PrintOut(FromPage, ToPage, Copies, Collate, Printer, PrintFileName)
```

C#

```
void PrintOut(  
    System.int FromPage,  
    System.int ToPage,  
    System.int Copies,  
    System.bool Collate,  
    System.string Printer,  
    System.string PrintFileName  
)
```

C++/CLI

```

void PrintOut(
& System.int FromPage,
& System.int ToPage,
& System.int Copies,
& System.bool Collate,
& System.String^ Printer,
& System.String^ PrintFileName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FromPage

ToPage

Copies

Collate

Printer

PrintFileName

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PrintOut](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

PrintOut2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PrintOut2 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::PrintOut3](#).

.NET Syntax

Visual Basic (Declaration)

```

Sub PrintOut2( _
    ByVal PageArray As System.Object, _
    ByVal Copies As System.Integer, _
    ByVal Collate As System.Boolean, _
    ByVal Printer As System.String, _
    ByVal PrintFileName As System.String _
)

```

Visual Basic (Usage)

```

Dim instance As IModelDocExtension
Dim PageArray As System.Object
Dim Copies As System.Integer
Dim Collate As System.Boolean
Dim Printer As System.String
Dim PrintFileName As System.String

```

```
instance.PrintOut2(PageArray, Copies, Collate, Printer, PrintFileName)
```

C#

```

void PrintOut2(
    System.object PageArray,
    System.int Copies,
    System.bool Collate,
    System.string Printer,

```



```
        System.string PrintFileName  
    )
```

C++/CLI

```
void PrintOut2(  
& System.Object^ PageArray,  
& System.int Copies,  
& System.bool Collate,  
& System.String^ Printer,  
& System.String^ PrintFileName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PageArray

Range of pages to print (see **Remarks**)

Copies

Number of copies to print

Collate

True to collate copies, false to not

Printer

Name of the printer queue (see **Remarks**)

PrintFileName

Name of file to print to instead of Printer

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PrintOut2](#).

Example

[Print Drawing and Save As PDF \(VBA\)](#)

[Print Drawing Document to File \(VBA\)](#)

Remarks

This method supports printing parts, assemblies, and both single and multisheet drawings.

PageArray

For a part or assembly, the PageArray argument is ignored. No dialogs or message boxes are displayed.

PageArray contains any number of pairs of values, each pair indicating the start page and end page of a range of pages to print. For example, to print sheets 1, 2, 3, 6, and 7 of a drawing, the array should contain 4 elements; 1, 3, 6, 7. This means to print pages 1-3 and 6-7. A range can be 5, 5, which means to print just page 5. If the array contains only one value, only that page is printed. If that one element is 0, then all sheets are printed.

If PageArray is an empty array, then all sheets are printed. Also, the array can contain just a one value, rather than an array, in which case only that page is printed, just like passing in an array containing only one value.

PrintFileName

To print to a file instead to a printer, set PrintFileName to a non-empty name.

If the PrintFileName is empty, then the document is printed to the printer specified in Printer. If that string is empty, then the document is printed to the default printer for this document. Use [IModelDoc2::Printer](#) to get or set this value.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDoc2::PrintDirect Method](#)

[IModelDoc2::PrintPreview Method](#)

[IModelDoc2::ClosePrintPreview Method](#)

[IModelDocExtension::IPrintOut2 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : PrintOut3 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::PrintOut4](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub PrintOut3( _  
    ByVal PageArray As System.Object, _  
    ByVal Copies As System.Integer, _  
    ByVal Collate As System.Boolean, _  
    ByVal Printer As System.String, _  
    ByVal PrintFileName As System.String, _  
    ByVal ConvertToHighQuality As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim PageArray As System.Object  
Dim Copies As System.Integer  
Dim Collate As System.Boolean  
Dim Printer As System.String  
Dim PrintFileName As System.String  
Dim ConvertToHighQuality As System.Boolean  
  
instance.PrintOut3(PageArray, Copies, Collate, Printer, PrintFileName, ConvertToHighQuality)
```

C#

```
void PrintOut3(  
    System.object PageArray,  
    System.int Copies,  
    System.bool Collate,  
    System.string Printer,  
    System.string PrintFileName,  
    System.bool ConvertToHighQuality  
)
```

C++/CLI

```
void PrintOut3(  
& System.Object^ PageArray,  
& System.int Copies,  
& System.bool Collate,  
& System.String^ Printer,  
& System.String^ PrintFileName,  
& System.bool ConvertToHighQuality  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PageArray
Range of pages to print (see **Remarks**)
Copies
Number of copies to print
Collate
True to collate copies, false to not
Printer
Name of the printer queue (see **Remarks**)
PrintFileName
Name of file to print to instead of Printer
ConvertToHighQuality
True to convert to high quality, false to leave as draft quality

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::PrintOut3](#).

Remarks

This method supports printing parts, assemblies, and both single and multisheet drawings.

PageArray

For a part or assembly, the PageArray argument is ignored. No dialogs or message boxes are displayed.

PageArray contains any number of pairs of values, each pair indicating the start page and end page of a range of pages to print. For example, to print sheets 1, 2, 3, 6, and 7 of a drawing, the array should contain 4 elements; 1, 3, 6, 7. This means to print pages 1-3 and 6-7. A range can be 5, 5, which means to print just page 5. If the array contains only one value, only that page is printed. If that one element is 0, then all sheets are printed.

If PageArray is an empty array, then all sheets are printed. Also, the array can contain just one value, rather than an array, in which case only that page is printed.

PrintFileName

To print to a file instead to a printer, set PrintFileName to a non-empty name.

If the PrintFileName is empty, then the document is printed to the printer specified in Printer. If that string is empty, then the document is printed to the default printer for this document. Use [IModelDoc2::Printer](#) to get or set this value.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::IPrintOut3 Method](#)
[IModelDoc2::PrintDirect Method](#)
[IModelDoc2::PrintPreview Method](#)
[IModelDoc2::ClosePrintPreview Method](#)
[IModelDocExtension::SaveAs Method](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0